

Algorithmic Advances for the
Design and Analysis of Being a Good Boy

A Dissertation
Presented to the Faculty of the Graduate School
of
Yale University
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Doctor of Philosophy

by
Handsome Dan

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Abstract

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Handsome Dan
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Since the beginning of recorded history, being a good boy is one of the most fundamental societal problems facing all canines. In this dissertation, we develop a mathematical framework for being a good boy and provide algorithms for optimal good boy behavior.

In the first chapter, we present a decision theoretic framework for modeling standard good boy tricks, including “sit”, “stay”, “paw”, and “down”. We show our framework is flexible enough to incorporate Bayesian priors based on standard exponential families. As an application of our framework, we demonstrate that the problem of learning to optimally walk your human is polynomial time solvable.

In the second chapter, we present a suite of algorithms for computing optimal good boy behavior. We show that, perhaps surprisingly, greedy algorithms fail to provide even a constant factor approximation. On the other hand, we develop a class of semidefinite programming based algorithms which allow for approximately optimal good boy behavior in polynomial time.

Lay Summary

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Perhaps the greatest paper of all is the walks we took along the way. This dissertation is dedicated to everyone who gave me a head pat during my time here at Yale.

Acknowledgment

First and foremost, I am eternally grateful to my academic advisor, Peter Salovey. In addition to the countless hours of stimulating discussion, the sweet bluegrass sounds of his bass have inspired many of the results in this dissertation. For this and more, I offer my sincerest thanks.

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Finally, thank you to my dearest family for their undying love: my mother, my father, and my 9 siblings in the litter. Your continued support and playtime has been the greatest gift of all.

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Chapter 1

Introduction

1.1 Motivation

The problem of good boy behavior has been well established in the literature. We follow the convention of Neyman [3] for the potential outcomes model. Moreover, we build on the work of Harshaw et al. [1] and Harshaw et al. [2] for design and analysis of experiments. We really cited these just so we'd have a non-empty bibliography.

In order to ensure that we have made proper use of the **figure** directory, we should create a figure. Figure 1.1 shows the optimal good boy, the original Handsome Dan. Finally, we end this motivation with nonsense text to fill it out.

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Figure 1.1: Optimal Good Boy

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Chapter 2

Good Boy Decision Theoretic Framework

2.1 Good boy behavior

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2.2 Formal theory

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Chapter 3

Efficient Good Boy Algorithms

3.1 Greedy algorithms fail

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3.2 Semdefinite programming approaches

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Appendix A

Proof of Mathematical Theorems

This appendix contains the proofs of all claims made in main body of the dissertation.

A.1 Proof of Theorem 1

The proof is left as an exercise to the reader.